

LIST OF EXPERIMENTS

1. Boyle's Law: This is an experiment to determine the relationship between the pressure and volume of a gas. It was first proposed by Robert Boyle in 1662.
2. Charles's Law: This is an experiment to determine the relationship between the volume and temperature of a gas. It was first proposed by Jacques Charles in 1787.
3. Avogadro's Law: This is an experiment to determine the relationship between the volume and number of moles of a gas. It was first proposed by Amedeo Avogadro in 1811.
4. Ideal Gas Law: This is an experiment to determine the relationship between the pressure, volume, temperature, and number of moles of a gas. It was first proposed by Emile Clapeyron in 1834.
5. Faraday's Law of Electrolysis: This is an experiment to determine the relationship between the quantity of a substance deposited at an electrode and the amount of electric current passed through an electrolyte. It was first proposed by Michael Faraday in 1833.
6. Ohm's Law: This is an experiment to determine the relationship between the voltage and current of an electrical circuit. It was first proposed by Georg Ohm in 1827.
7. Bernoulli's Principle: This is an experiment to determine the relationship between the pressure and velocity of a fluid. It was first proposed by Daniel Bernoulli in 1738.
8. Pascal's Law: This is an experiment to determine the relationship between the pressure and height of a fluid. It was first proposed by Blaise Pascal in 1647.

Experiments to be Performed

1. Observing the effects of different concentrations of a reactant on the rate of a reaction.
2. Investigating the effect of temperature on the rate of a reaction.
3. Measuring the rate of a reaction over time.
4. Investigating the effect of a catalyst on the rate of a reaction.
5. Measuring the rate of a reaction at different pH levels.
6. Investigating the effect of light on the rate of a reaction.
7. Examining the effect of a change in surface area on the rate of a reaction.
8. Investigating the effect of pressure on the rate of a reaction.
9. Examining the effect of a change in concentration on the rate of a reaction.
10. Investigating the effect of a change in the nature of the reactants on the rate of a reaction.

(A) Hardware based experiments

1. Hardware based experiments involve the use of physical hardware to measure and record data.
2. This type of experiment is typically used to study physical phenomena, such as the behavior of a system or the properties of a material.
3. Common hardware used in experiments include sensors, transducers, and other measuring devices, as well as computers, microcontrollers, and other data-logging systems.
4. Experiments can be conducted in a laboratory setting or in the field.
5. In order to ensure accuracy and reliability, experiments must be carefully designed, taking into account factors such as the type of hardware being used and the environment in which the experiment is being conducted.
6. Data collected from hardware based experiments must be carefully analyzed in order to draw meaningful conclusions.

1. Verification of Kirchhoff's Laws

1. Kirchhoff's Voltage Law (KVL): This law states that the sum of all the voltages around a closed loop in a circuit is equal to zero. This can be verified by applying the law of conservation of energy.
2. Kirchhoff's Current Law (KCL): This law states that the sum of all the currents entering a node in a circuit is equal to the sum of all the currents leaving that node. This can be verified by applying the law of conservation of charge.
3. Kirchhoff's Rules: These are two equations, KVL and KCL, which can be used to analyze electrical circuits. They can be verified by applying the laws of conservation of energy and charge.

2. Measurement of power and power factor in a single phase ac series inductive circuit and study improvement of power factor using capacitor

1. Power factor is the ratio between the real power and the apparent power in an AC circuit.
2. In a single-phase AC series inductive circuit, the power factor is always less than one.
3. The power factor in a single-phase AC series inductive circuit can be improved by adding a capacitor in parallel with the load.
4. The value of the capacitor should be calculated based on the load current, the line voltage, and the power factor desired.
5. The power factor can be measured by using a wattmeter or a power factor meter.

6. The wattmeter is used to measure the real power in the circuit, while the power factor meter is used to measure the power factor.
7. The power factor can be improved by adding a capacitor in parallel with the load. The capacitor should be sized based on the load current, line voltage, and desired power factor.

3. Study of phenomenon of resonance in RLC series circuit and obtain resonant frequency

1. Resonance is the phenomenon of increased amplitude of oscillations in an RLC series circuit when the frequency of the applied voltage is equal to the resonant frequency of the circuit.
2. The resonant frequency of an RLC series circuit is determined by the values of the inductance (L), capacitance (C) and resistance (R) in the circuit.
3. The resonant frequency of an RLC series circuit is given by the formula:

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

4. The resonant frequency of an RLC series circuit can be calculated by substituting the values of the inductance, capacitance and resistance into the above formula.
5. The resonant frequency of an RLC series circuit can also be determined experimentally by measuring the frequency at which the amplitude of oscillations in the circuit is maximum.

4. Connection and Measurement of Power Consumption of a Fluorescent Lamp (Tube Light)

1. A fluorescent lamp (tube light) is an electric light that uses electricity to excite mercury vapor in a tube, which produces ultraviolet light.
2. This ultraviolet light then causes a phosphor to fluoresce, producing visible light.
3. Fluorescent lamps are more efficient than incandescent lamps, and can last up to 10 times longer.
4. To connect a fluorescent lamp, the lamp is inserted into the holder and the holder is connected to the power supply.
5. The power consumption of a fluorescent lamp can be measured using a wattmeter.
6. The wattmeter is connected in series with the lamp, and the power consumption of the lamp can be read from the wattmeter.

5. Measurement of power in 3-phase circuit by two-wattmeter method and determination of its power factor for star as well as delta connected load

1. In a 3-phase system, the power can be measured by two-wattmeter method.
2. This method is based on the measurement of power in each phase of the system.
3. The wattmeter used in the two-wattmeter method is connected in the star or delta connection of the load.
4. The wattmeter measures the power in each phase and the total power of the system is calculated by adding the power in each phase.
5. The power factor of the system can also be determined using the two-wattmeter method.
6. The power factor of the system is calculated by dividing the total power in the system by the total apparent power in the system.
7. The power factor of the system can be calculated for both star and delta connected loads.

6. Determination of parameters of ac single phase series RLC circuit

1. The ac single phase series RLC circuit consists of a resistor (R), an inductor (L) and a capacitor (C) connected in series.
2. The circuit can be used to measure the parameters of an ac signal, such as the frequency, the peak voltage, the peak current and the phase angle between the voltage and the current.
3. The frequency of the ac signal is determined by the ratio of the inductive reactance (XL) to the capacitive reactance (XC).
4. The peak voltage is determined by the total impedance of the circuit (ZT).
5. The peak current is determined by the total impedance of the circuit and the peak voltage.
6. The phase angle between the voltage and the current is determined by the difference between the inductive reactance and the capacitive reactance.

7. Determination of (i) Voltage ratio (ii) polarity and (iii) efficiency by load test of a single phase Transformer

1. Voltage ratio: The voltage ratio of a single phase transformer can be determined by connecting it to a variable voltage supply and measuring the voltage at the primary and secondary windings.
2. Polarity: The polarity of the transformer can be determined by connecting it to a variable voltage supply and measuring the voltage across the primary and secondary windings. The polarity of the transformer can also be determined by connecting it to an ammeter and observing the direction of the current flow.

3. Efficiency: The efficiency of a single phase transformer can be determined by connecting it to a variable voltage supply and measuring the input and output power. The efficiency is then calculated as the output power divided by the input power.

8. Determination of efficiency of a dc shunt motor by load test

1. A dc shunt motor is an electric motor that works on direct current (DC) and has its field windings connected in parallel to the armature windings.
2. The efficiency of a dc shunt motor can be determined by a load test.
3. During a load test, the motor is connected to a DC power supply and its speed and torque are measured while the load is varied.
4. The efficiency of the motor is calculated by comparing the input power to the output power.
5. The input power is calculated by multiplying the voltage, current and power factor of the motor.
6. The output power is calculated by multiplying the torque and speed of the motor.
7. The efficiency of the motor is calculated by dividing the output power by the input power.
8. The efficiency of the motor can be improved by using a higher voltage power supply and by increasing the power factor of the motor.

9. To study running and speed reversal of a three phase induction motor and record speed in both directions

1. A three phase induction motor is an AC motor in which the stator windings are connected to a three phase AC power supply and the rotor is given a rotating magnetic field to produce torque.
2. The speed of the motor is determined by the frequency of the AC power supply, and the direction of rotation is determined by the phase sequence of the supply.
3. To study the running and speed reversal of a three phase induction motor, the following steps should be followed:
 - a. Connect the stator windings of the motor to a three phase AC power supply.
 - b. Change the phase sequence of the AC power supply to reverse the direction of rotation.
 - c. Measure the speed of the motor in both directions and record the results.
 - d. Calculate the efficiency of the motor in both directions.
4. By studying the running and speed reversal of a three phase induction motor, it is possible to gain insight into the performance of the motor and to identify any problems or areas for improvement.

10. Demonstration of cut-out sections of machines:

- **DC Machine:** A DC Machine is an electrical machine that converts electrical energy into mechanical energy. It consists of two main parts: the stator and the rotor. The stator consists of a field winding and a commutator, while the rotor consists of an armature winding. The armature winding is connected to the commutator, which is responsible for reversing the direction of the current in the armature winding.
- **Three Phase Induction Machine:** A three phase induction machine is a type of electrical machine that uses three-phase alternating current to produce a rotating magnetic field. It consists of a stator, which is made up of three windings, and a rotor, which is made up of a single winding. The stator windings are connected to a three-phase power supply, while the rotor winding is connected to a load.
- **Single-Phase Induction Machine:** A single-phase induction machine is an electrical machine that uses a single-phase alternating current to produce a rotating magnetic field. It consists of a stator, which is made up of two windings, and a rotor, which is made up of a single winding. The stator windings are connected to a single-phase power supply, while the rotor winding is connected to a load.
- **Synchronous Machine:** A synchronous machine is an electrical machine that produces a rotating magnetic field in synchronism with the frequency of the power supply. It consists of a stator, which is made up of two windings, and a rotor, which is made up of a single winding. The stator windings are connected to a three-phase power supply, while the rotor winding is connected to a load.

(B) Experiments available on virtual lab

1. Virtual labs can provide users with a wide range of experiments and activities, such as:
 - **Physics:** simulations of physical phenomena, such as light refraction, electrical circuits and the Doppler effect.
 - **Chemistry:** simulations of chemical reactions, such as the combustion of hydrocarbons and the reaction of acids and bases.
 - **Biology:** simulations of biological processes, such as cell division and photosynthesis.
2. Virtual labs can also provide users with access to laboratory equipment, such as microscopes and spectrometers, which can be used to conduct experiments.
3. Virtual labs can provide users with access to data analysis tools, such as graphing calculators and statistical software.
4. Virtual labs can also provide users with access to simulations of real-world experiments, such as the famous Millikan Oil Drop experiment.
5. By using virtual labs, users can gain experience with laboratory techniques, data analysis and scientific inquiry without the need for expensive

equipment or hazardous materials.

1. Kirchhoff's Laws

1. **Kirchhoff's Current Law (KCL):** The algebraic sum of currents entering and leaving any node (junction) is zero.
2. **Kirchhoff's Voltage Law (KVL):** The algebraic sum of the voltages around any closed loop is zero.
3. **Kirchhoff's Junction Rule (KJR):** The sum of currents entering a junction is equal to the sum of currents leaving the junction.
4. **Kirchhoff's Loop Rule (KLR):** The algebraic sum of the voltage drops in any closed loop is equal to the algebraic sum of the emf's in the same loop.

Virtual Lab

1. Virtual lab is a web-based platform that provides access to a wide range of experiments and simulations.
2. It is designed to help students learn and understand the concepts of science, engineering, and technology.
3. The virtual lab provides an interactive environment where users can explore, experiment, and learn.
4. It is a great tool for teachers to use in their classrooms.
5. It is also a great way for students to practice and apply their knowledge.
6. The virtual lab link provided is <http://vlab.amrita.edu/?sub=3&brch=75&sim=217&cnt=2>. It provides access to a variety of experiments and simulations related to engineering and technology.
7. Users can explore, experiment, and learn about the topics provided in the virtual lab.
8. The virtual lab allows users to interact with the experiments and simulations in a realistic and engaging way.
9. The virtual lab also provides users with the opportunity to collaborate with others and to share their results.
10. The virtual lab is a great tool for learning and understanding the concepts of science, engineering, and technology.

2. Thevenin Theorem

- Thevenin Theorem states that any linear, active, two-terminal network can be reduced to an equivalent circuit consisting of a single voltage source in series with a single impedance.
- The equivalent voltage source is the open-circuit voltage at the two terminals and the equivalent impedance is the impedance looking into the two terminals when all sources are turned off.

- The Thevenin equivalent circuit can be used to find the current through any load connected between the two terminals.
- Thevenin Theorem can be used to simplify complex electrical networks and reduce the number of components needed to analyze a circuit.
- Thevenin Theorem can also be used to calculate the voltage across a component in a circuit when the current through the component is known.

Virtual Lab

Virtual labs are a great way to experiment with different technologies and gain hands-on experience in a safe and controlled environment. The following link provides access to a virtual lab:

- **URL:** <https://vlab.amrita.edu/?sub=1&brch=75&sim=313&cnt=1>

This virtual lab provides access to a range of simulations and experiments related to electrical engineering. It is an excellent resource for students to gain a better understanding of the principles and applications of electrical engineering.

The virtual lab includes the following features:

- **Simulation:** A range of simulations related to electrical engineering topics.
- **Experiments:** Experiments that can be conducted in a safe and controlled environment.
- **Data Analysis:** Analyze the data collected from the experiments.
- **Quizzes:** Assess your understanding of the topics with quizzes.
- **Tutorials:** Step-by-step tutorials to help you understand the concepts.
- **Assignments:** Assignments to help you practice and apply the concepts.

3. RLC Series Resonance

1. RLC Series Resonance is a type of circuit that consists of three components: a resistor, an inductor, and a capacitor.
2. When an alternating current (AC) is applied to the circuit, the components interact to form a resonant frequency, which is the frequency at which the circuit will oscillate.
3. The resonant frequency is determined by the values of the resistor, inductor, and capacitor.
4. The resonant frequency of an RLC circuit can be calculated using the formula:

$$f = 1 / (2 \sqrt{LC})$$

where f is the resonant frequency, L is the inductance, and C is the capacitance.

5. At the resonant frequency, the current in the circuit is at a maximum, and the voltage across the components is at a minimum.
6. RLC series resonance

circuits are used in a variety of applications, such as filters, oscillators, and amplifiers.

Virtual Lab

1. Virtual Lab is an online platform that provides access to a range of interactive experiments and simulations in various disciplines.
2. It enables students to gain hands-on experience in a safe and secure virtual environment.
3. The Virtual Lab platform is developed by Amrita University, India, and provides access to experiments and simulations in various disciplines such as Electrical and Electronics, Computer Science, Physics, and Chemistry.
4. It is a great resource for students to gain a better understanding of the concepts and principles they are learning in their respective courses.
5. The platform is easy to use and provides a user-friendly interface.
6. It also provides a variety of tools and resources to help students learn more effectively.
7. The Virtual Lab platform is free to use and provides access to a wide range of experiments and simulations.

4. Measurement of power in 3-phase circuit by two wattmeter method and determination of its power factor for star as well as delta connected load.

1. Power in three-phase circuit can be measured using two wattmeter method.
2. For star connected load, the two wattmeters measure the power in two of the three phases.
3. For delta connected load, the two wattmeters measure the power in two of the three lines.
4. The power factor of the circuit is determined by the ratio of the power measured by the two wattmeters.
5. The power factor is the ratio of the active power to the apparent power.
6. The active power is the power measured by the wattmeters, while the apparent power is the product of the voltage and current.

Measurement in Virtual Lab

1. Measurement is the process of obtaining a numerical or qualitative value of a physical quantity.
2. Measurement is used in many fields such as science, engineering, economics and medicine.
3. The virtual lab link provided allows users to explore the different types of measurements and their applications.
4. The link provides information on the various instruments used for measurement such as voltmeters, ammeters, wattmeters, and more.

5. It also provides an in-depth understanding of the principles of measurement and the various methods used to measure physical quantities such as length, mass, time, temperature, pressure, and more.
6. The link also provides interactive simulations of the measurement process and allows users to practice their measurement skills.
7. The virtual lab link is a great resource for students and professionals to learn about the fundamentals of measurement and its applications.

5. Determination of parameters of ac single phase series RLC circuit

1. The parameters of a single-phase series RLC circuit can be determined by applying the following equations:

- Inductive Reactance:

$$X_L = 2\pi fL$$

- Capacitive Reactance:

$$X_C = \frac{1}{2\pi fC}$$

- Total Reactance:

$$X_T = X_L + X_C$$

- Impedance:

$$Z = \sqrt{R^2 + X_T^2}$$

- Phase Angle:

$$\theta = \tan^{-1} \frac{X_T}{R}$$

- Apparent Power:

$$S = V \times I$$

- Active Power:

$$P = V \times I \times \cos \theta$$

- Reactive Power:

$$Q = V \times I \times \sin \theta$$

2. The parameters of a series RLC circuit can also be determined by using the phasor diagram. The phasor diagram is a graphical representation of the circuit parameters where the magnitude and phase angle of each component are represented by vectors. The magnitude of the vectors is determined by the equations mentioned above and the phase angle is determined by the relative position of the components in the circuit.
3. The parameters of a series RLC circuit can also be determined by using the mesh analysis method. In this method, the mesh currents are determined by applying Kirchhoff's current law and then the circuit parameters can be determined by applying the equations mentioned above.

Introduction to Virtual Lab

- Virtual Lab is an online platform that provides access to simulations and experiments in various fields of science and engineering.
- It is developed by Amrita University and allows students to perform experiments and simulations in a virtual environment.
- The platform is designed to be interactive and user-friendly, making it easy for students to understand the concepts and principles behind the experiments.
- The Virtual Lab platform provides access to a wide range of simulations and experiments in fields such as electronics, electrical engineering, computer science, physics, chemistry, and biology.
- The simulations and experiments can be accessed through the Virtual Lab link provided above.
- The experiments and simulations are designed to help students understand the concepts in an interactive and engaging manner.
- The platform also provides access to tutorials and other resources to help students understand the experiments better.
- The Virtual Lab platform is free to use and can be accessed from any device with an internet connection.

6. To observe the B-H loop of a ferromagnetic material in CRO

1. A ferromagnetic material is a material that is strongly affected by a magnetic field.
2. The B-H loop is a graph that represents the relationship between the magnetizing field (H) and the magnetic flux density (B) of a ferromagnetic material.
3. The B-H loop is typically constructed by applying a range of increasing and decreasing magnetic fields to a ferromagnetic material and recording the corresponding magnetic flux density.
4. The B-H loop can be observed using a Cathode Ray Oscilloscope (CRO).
5. To observe the B-H loop of a ferromagnetic material using a CRO, the ferromagnetic material is placed in the magnetic field of the CRO.
6. The CRO is then used to measure the magnetic field and the corresponding magnetic flux density.
7. The resulting data is plotted on a graph to form the B-H loop.
8. The shape of the B-H loop can be used to determine the magnetic properties of the ferromagnetic material.

Virtual Labs

Virtual labs are a type of online laboratory environment that allow users to access and interact with a range of virtual experiments. Virtual labs are typically used in educational contexts to supplement or replace physical lab experiments.

- Virtual labs provide a safe, controlled environment for students to explore

and experiment with scientific concepts.

- They can be used to simulate real-world experiments, allowing students to explore the same concepts without the need for expensive or dangerous equipment.
- Virtual labs are often used in distance learning courses, allowing students to access the same experiments from any location.
- Virtual labs are also used to provide students with access to experiments that may not be available in their local area.
- Virtual labs can be used to supplement traditional lab experiments, giving students the opportunity to explore a wider range of topics.

7. Determination of the Efficiency of a DC Motor by Loss Summation Method (Swinburne's Test)

1. The Swinburne's Test is a method of determining the efficiency of a DC motor by summing up the losses.
2. The efficiency of the DC motor is calculated by measuring the power input to the motor and the power output from the motor.
3. The losses in the motor can be divided into two categories: losses due to friction and losses due to electrical resistance.
4. The losses due to friction are calculated by measuring the torque and speed of the motor.
5. The losses due to electrical resistance are calculated by measuring the current and voltage of the motor.
6. The total losses are calculated by summing up the losses due to friction and the losses due to electrical resistance.
7. The efficiency of the motor is calculated by subtracting the total losses from the power input to the motor.
8. The virtual lab link provided above provides an interactive simulation of the Swinburne's Test. It allows the user to measure the torque, speed, current and voltage of the motor and calculate the efficiency of the motor.

Course Outcomes:

- Understand and explain the fundamentals of software engineering
- Develop and evaluate software solutions using appropriate development methodologies
- Analyze and design software solutions using appropriate software engineering tools and techniques
- Implement and test software solutions using appropriate software engineering tools and techniques
- Evaluate the impact of software engineering on society and the environment
- Develop and demonstrate an understanding of professional, ethical, legal, security and social issues related to software engineering

Course Outcome (CO) Bloom's Level

- **Remember:** Students will recall or recognize facts, terms, concepts, and answers.
- **Understand:** Students will demonstrate understanding of facts, terms, concepts, and answers.
- **Apply:** Students will use information to solve problems in new or familiar situations.
- **Analyze:** Students will break down information into parts and recognize relationships among the parts.
- **Evaluate:** Students will make judgments about the value of ideas or materials.
- **Create:** Students will put elements together to form a coherent or functional whole; to generate a new pattern or structure.

At the end of this course, the students should be able to:

- Understand the fundamentals of computer programming
- Identify the main components of a computer system
- Develop algorithms to solve problems
- Design and implement programs in a programming language
- Analyze the performance of programs
- Use debugging techniques to identify and fix errors in programs
- Demonstrate proficiency in the use of software development tools
- Create user-friendly interfaces for programs
- Utilize best practices for software development

CO 1 Conduct experiments illustrating the application of KVL/KCL and network theorems to DC electrical circuits. K3

1. Kirchhoff's Voltage Law (KVL) states that the sum of the voltages around a closed loop in a circuit is equal to zero.
2. Kirchhoff's Current Law (KCL) states that the sum of the currents entering and exiting a node in a circuit is equal to zero.
3. Network Theorems provide a set of equations that allow us to calculate the voltage, current, and power of a circuit.
4. Ohm's Law states that the voltage across a resistor is proportional to the current flowing through it.
5. Thevenin's Theorem states that any linear circuit can be replaced by an equivalent circuit consisting of a voltage source in series with a resistor.
6. Norton's Theorem states that any linear circuit can be replaced by an equivalent circuit consisting of a current source in parallel with a resistor.
7. Superposition Theorem states that the response of a linear circuit to multiple sources can be determined by considering each source separately and then adding the responses together.
8. Maximum Power Transfer Theorem states that the maximum power is transferred from a source to a load when the load resistance is equal to

the source resistance.

CO 2 Demonstrate the behavior of AC circuits connected to single phase AC supply and measure power in single phase as well as three phase electrical circuits. K4

1. AC circuits are electrical circuits that use alternating current. This type of current flows in one direction, then reverses and flows in the opposite direction.
2. AC circuits are typically connected to a single-phase AC supply, which is a type of alternating current power source. This power source is often used in residential and commercial applications.
3. In order to measure the power of an AC circuit, it is necessary to measure both the voltage and the current. This can be done using a multimeter.
4. Power in single-phase AC circuits can be calculated using the formula $P = VI$, where P is the power, V is the voltage, and I is the current.
5. Three-phase AC circuits are more complex than single-phase AC circuits. In order to measure the power of a three-phase AC circuit, it is necessary to measure the voltage and current in each of the three phases.
6. Power in three-phase AC circuits can be calculated using the formula $P = 3VI\cos\Phi$, where P is the power, V is the voltage, I is the current, and Φ is the phase angle.

CO 3 Perform experiment illustrating BH curve of magnetic materials. K3

1. BH curves are graphical representations of the relationship between magnetic flux density (B) and magnetic field strength (H).
2. BH curves can be used to measure the magnetic properties of materials, such as their coercivity, remanence, and saturation magnetization.
3. The experiment involves creating a BH curve by plotting the magnetic flux density (B) against the magnetic field strength (H).
4. The experiment requires a source of magnetic field, such as an electromagnet or a permanent magnet, and a device to measure the magnetic flux density, such as a Hall effect sensor.
5. The experiment is conducted by increasing the magnetic field strength (H) from 0 to its maximum value, while measuring the corresponding magnetic flux density (B) at each step.
6. The resulting BH curve can be used to measure the coercivity, remanence, and saturation magnetization of the material being tested.

CO 4 Calculate efficiency of a single phase transformer and DC machine. K4

1. Efficiency of a single phase transformer is the ratio of output power to input power, expressed as a percentage.
2. The efficiency of a DC machine can be calculated by dividing the output power by the input power and then multiplying the result by 100.
3. The single phase transformer efficiency is affected by the load, the voltage, the frequency, the temperature, and the age of the transformer.
4. The DC machine efficiency is affected by the load, the speed, the temperature, and the age of the machine.
5. In order to maximize the efficiency of a single phase transformer, it is important to ensure that the load is balanced, the voltage is correct, the frequency is correct, the temperature is correct, and the transformer is not too old.
6. In order to maximize the efficiency of a DC machine, it is important to ensure that the load is balanced, the speed is correct, the temperature is correct, and the machine is not too old.

CO 5 Perform experiments on speed measurement and reversal of direction of three phase induction motor and Identify the type of DC and AC machines based on their construction. K4

1. Speed Measurement of Three Phase Induction Motor:
 - The speed of a three phase induction motor can be measured using a tachometer.
 - A tachometer is a device that measures the speed of a rotating shaft.
 - The tachometer is connected to the motor shaft and the speed is measured in revolutions per minute (RPM).
2. Reversal of Direction of Three Phase Induction Motor:
 - The direction of rotation of a three phase induction motor can be changed by reversing the sequence of the 3-phase supply.
 - The direction of rotation of the motor can also be changed by interchanging any two of the three supply leads.
3. Types of DC Machines:
 - DC machines can be classified into two main types:
 - DC generators
 - DC motors
4. Types of AC Machines:
 - AC machines can be classified into two main types:
 - AC generators
 - AC motors

K1 – Remember

- Remembering is the ability to recall and recall information.

- It involves the use of memory strategies such as repetition, organization, and visualization.
- Remembering is important for understanding concepts and retaining knowledge.

K2 – Understand

- Understanding is the ability to comprehend and interpret information.
- It involves the use of higher-order thinking skills such as analysis, synthesis, and evaluation.
- Understanding is important for problem solving and making decisions.

K3 – Apply

- Applying is the ability to use knowledge and skills in a new context.
- It involves the use of practical skills such as problem solving and critical thinking.
- Applying is important for developing new solutions and adapting to changing environments.

K4 – Analyze

- Analyzing is the ability to break down information into its component parts.
- It involves the use of analytical skills such as comparison, classification, and deduction.
- Analyzing is important for understanding complex concepts and making sense of data.

K5 – Evaluate

- Evaluating is the ability to assess the quality of information.
- It involves the use of evaluation skills such as judgment, reflection, and comparison.
- Evaluating is important for making informed decisions and determining the accuracy of information.

K6 – Create

- Creating is the ability to generate new ideas and solutions.
- It involves the use of creative skills such as imagination, innovation, and experimentation.
- Creating is important for developing new products and services and advancing the state of knowledge.